Decomposition methods for two-stage stochastic programming

Yiping Fang, Professor

Chair on Risk and Resilience of Complex Systems,

Laboratoire Génie Industriel, CentraleSupélec, Université Paris Saclay

Yiping.fang@centralesupelec.fr

Outline

- Benders decomposition/L-Shaped method
- Lagrangian dual decomposition

Production planning example

$$
\begin{cases}
\text{min} & 2x_1 + 3x_2 + \mathbb{E}[Q(x,\xi)] \\
\text{s.t.} & x_1 + x_2 \leq 100 \\
 & x_1, x_2 \geq 0
\end{cases}
$$

$$
Q(x,\xi) := \begin{cases}
\text{min} & 7y_1 + 12y_2 \\
\text{s.t.} & y_1 \geq h_1 - (\alpha x_1 + 6x_2) \\
 & y_2 \geq h_2 - (3x_1 + \beta x_2) \\
 & y_1, y_2 \geq 0
\end{cases}
$$

- Complete fixed recourse

The SAA model

$$
\begin{aligned}
\mathrm{Z}^\star = \text{min} \quad & 2x_1 + 3x_2 + \sum_s p_s[7y_1(\boldsymbol{\xi}_s) + 12y_2(\boldsymbol{\xi}_s)] \\
\text{s.t.} \quad & x_1 + x_2 \leq 100 \\
& x_1, x_2 \geq 0 \\
& \left.\begin{aligned}
&\alpha(\boldsymbol{\xi}_s)x_1 + 6x_2 + y_1(\boldsymbol{\xi}_s) \geq h_1(\boldsymbol{\xi}_s) \\
&3x_1 + \beta(\boldsymbol{\xi}_s)x_2 + y_2(\boldsymbol{\xi}_s) \geq h_2(\boldsymbol{\xi}_s) \\
&y_1(\boldsymbol{\xi}_s), y_2(\boldsymbol{\xi}_s) \geq 0
\end{aligned}\right\} \forall s
\end{aligned}
$$

- A deterministic LP, but potentially many constraints and 2nd-stage variables
- MIP when 1st stage variables are discrete/binary
- L-shaped structure of the constraints

The SAA model

$$\begin{array}{cccc}
\boxed{A} & & & \\
\boxed{T_1} & \boxed{W} & & \\
\boxed{T_2} & & \boxed{W} & \\
\vdots & & & \\
\boxed{T_N} & \dots & & \boxed{W}
\end{array}$$

Context for L-shaped method

- Entire problem is difficult to solve, due to
  - → The linking constraints
  - → If x is fixed, the 2nd-stage problem is relatively easy
  - → Potentially too many scenarios (leads to too many constraints and variables)

Dual of 2nd stage problem

$$\begin{cases} \text{min} & 2x_1 + 3x_2 + \boxed{\sum_s p_s Q(x, \xi_s)} \rightarrow \mathcal{Q}(x) \\ \text{s.t.} & x_1 + x_2 \leq 100 \\ & x_1, x_2 \geq 0 \end{cases}$$

$$Q(x,\xi) \overset{\text{primal}}{=} \begin{cases} \text{min} & 7y_1 + 12y_2 \\ \text{s.t.} & y_1 \geq h_1 - (\alpha x_1 + 6x_2) \\ & y_2 \geq h_2 - (3x_1 + \beta x_2) \\ & y_1, y_2 \geq 0 \end{cases}$$

$$\overset{\text{dual}}{=} \begin{cases} \text{max} & [h_1 - (\alpha x_1 + 6x_2)]\pi_1 + [h_2 - (3x_1 + \beta x_2)]\pi_2 \\ \text{s.t.} & \pi_1 \leq 7, \\ & \pi_2 \leq 12, \\ & \pi_1, \pi_2 \geq 0. \end{cases}$$

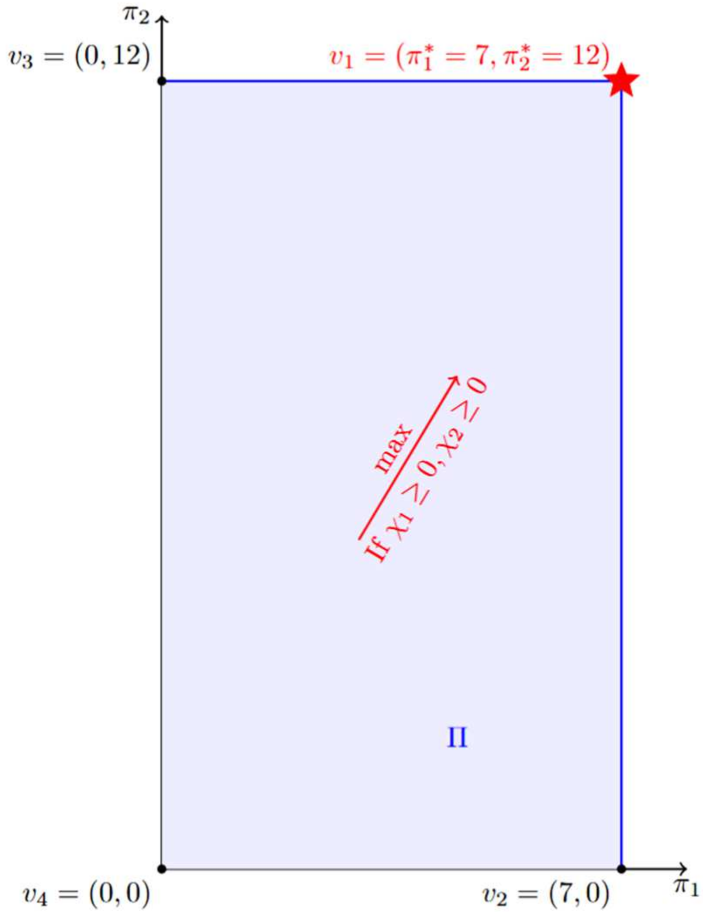
Dual of 2nd stage problem

$$Q(x,\xi) \overset{\text{dual}}{=} \begin{cases} \max & \underbrace{[h_1 - (\alpha x_1 + 6x_2)]}_{\chi_1}\pi_1 + \underbrace{[h_2 - (3x_1 + \beta x_2)]}_{\chi_2}\pi_2 \\ \text{s.t.} & (\pi_1, \pi_2) \in \Pi = \{\pi \in \mathbb{R}^2 : 0 \le \pi_1 \le 7, 0 \le \pi_2 \le 12\} \end{cases}$$

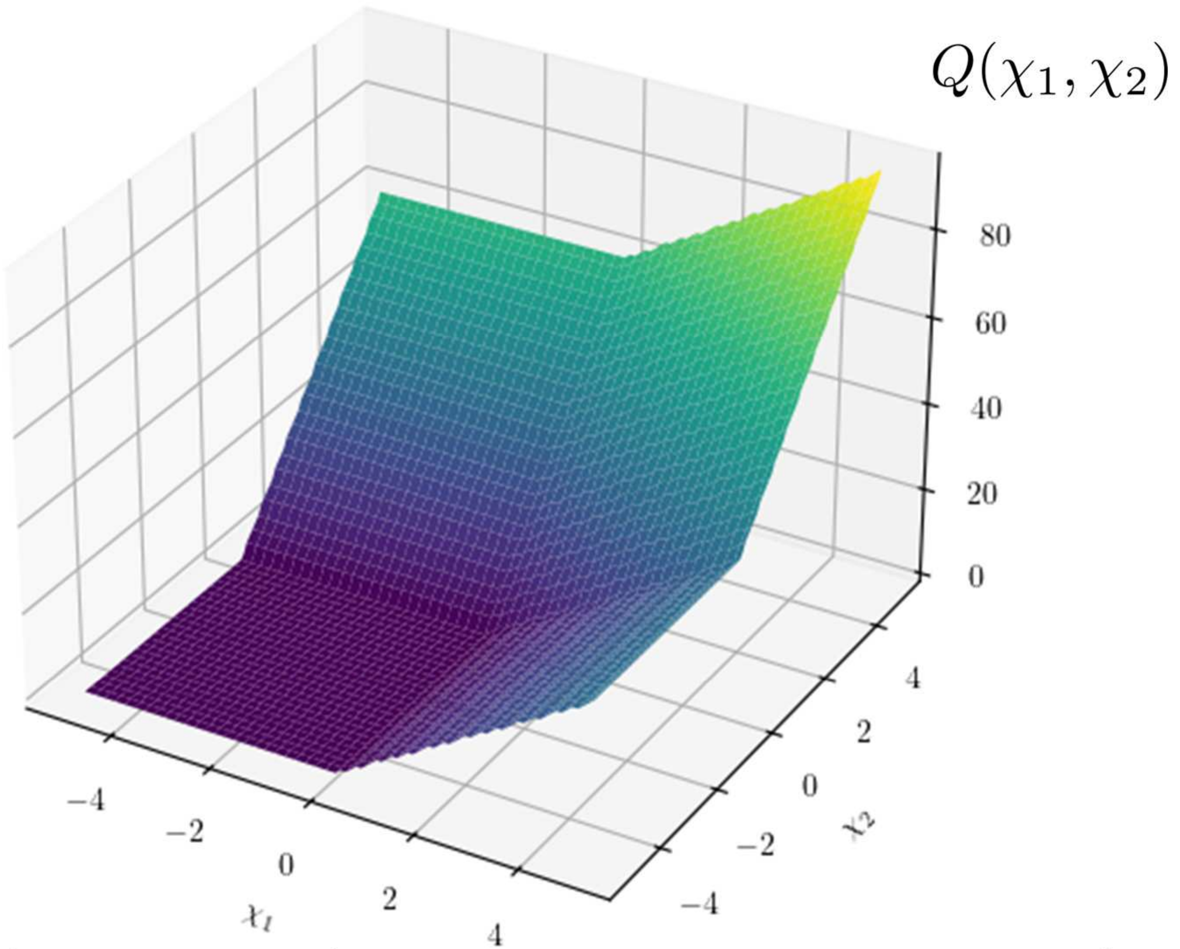
- Max of affine functions of x, take over a polyhedral set

$$Q(x,\xi) = \begin{cases} 7\chi_1 + 12\chi_2 & \text{if } \chi_1 \ge 0, \chi_2 \ge 0 \\ 7\chi_1 & \text{if } \chi_1 \ge 0, \chi_2 \le 0 \\ 12\chi_2 & \text{if } \chi_1 \le 0, \chi_2 \ge 0 \\ 0 & \text{if } \chi_1 \le 0, \chi_2 \le 0 \end{cases}$$

- Q is a piece-wise linear function in x for any given ξ
- $\mathcal{Q}$ is a piece-wise linear function in x

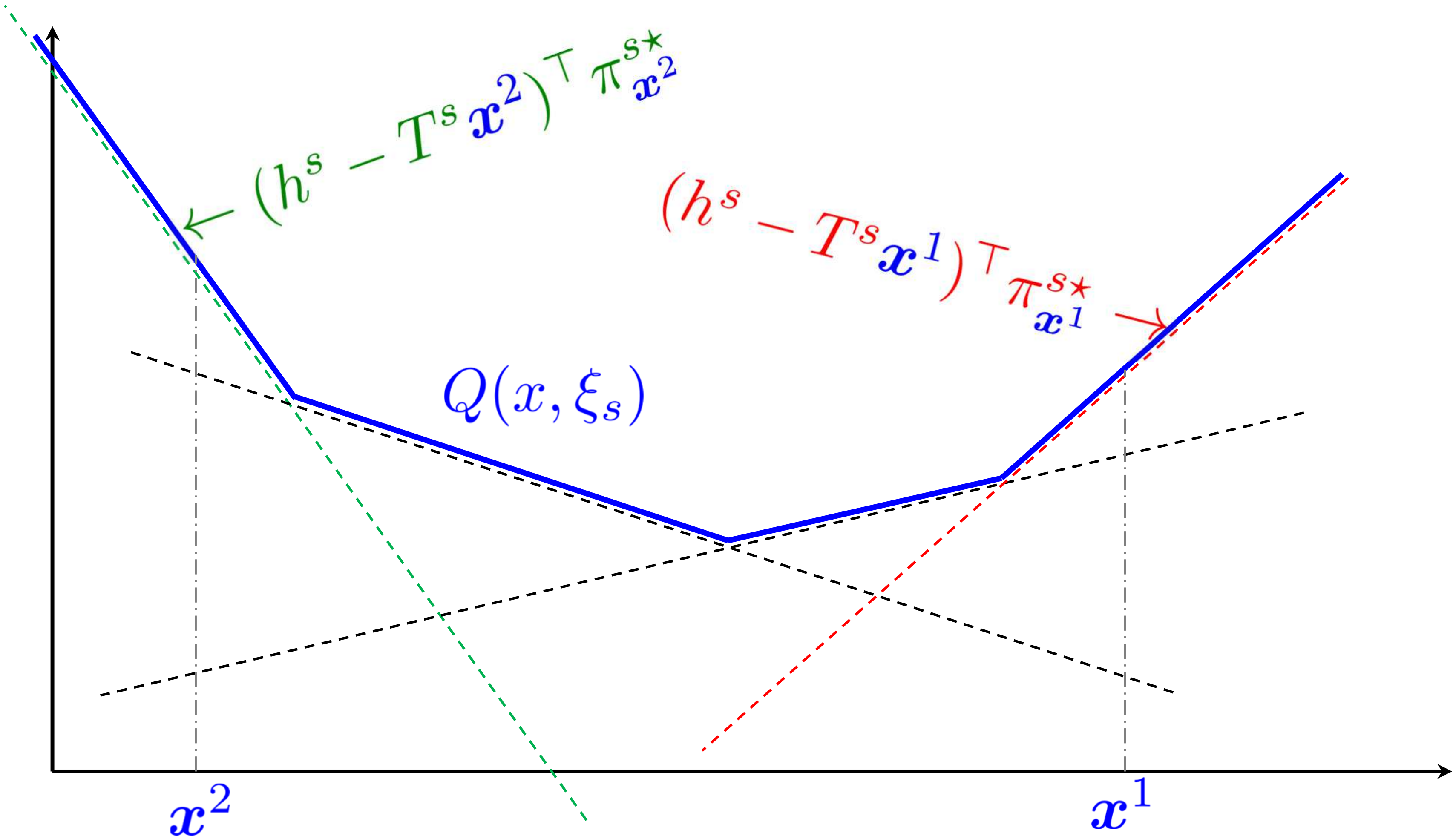


Second-stage value function $Q(x, \xi)$



$\chi_1 = h_1 - (\alpha x_1 + 6x_2)$ $\chi_2 = h_2 - (3x_1 + \beta x_2)$

$$Q(x,\xi_s) = \begin{cases} \min & \theta_s \\ \text{s.t.} & \theta_s \geq (h^s - T^s x)^\top \pi^s,\ \forall \pi^s \in \mathbf{XP}(\Pi_s) \end{cases}$$



Supporting hyperplane

- In the production planning example

$$\Pi_s = \Pi = \{\pi \in \mathbb{R}^2 : 0 \leq \pi_1 \leq 7, 0 \leq \pi_2 \leq 12\}$$

$$\mathbf{XP}(\Pi_s) = \{(0,0), (7,0), (0,12), (7,12)\}$$

$$Q(x, \xi_s) = \begin{cases} \text{min} & \theta_s \\ \text{s.t.} & \theta_s \geq [h_1^s - (\alpha^s x_1 + 6x_2)]\pi_1 + \\ & \qquad [h_2^s - (3x_1 + \beta^s x_2)]\pi_2 \\ & \qquad \forall (\pi_1, \pi_2) \in \mathbf{XP}(\Pi_s) \end{cases}$$

Reformulation and insights

$$
Z^\star = \begin{cases}
\min\limits_{x,\theta} & 2x_1 + 3x_2 + \sum_s p_s\theta_s \\
\text{s.t.} & x_1 + x_2 \le 100 \\
& x_1, x_2 \ge 0 \\
& \theta_s \ge [h_1^s - (\alpha^s x_1 + 6x_2)]\pi_1 + \\
& \qquad [h_2^s - (3x_1 + \beta^s x_2)]\pi_2 \\
& \qquad \forall(\pi_1, \pi_2) \in \mathbf{XP}(\Pi_s), \forall s
\end{cases}
$$

- 2nd stage variables replaced by the dual extreme points, which can be found independently for each scenario

Reformulation and insights

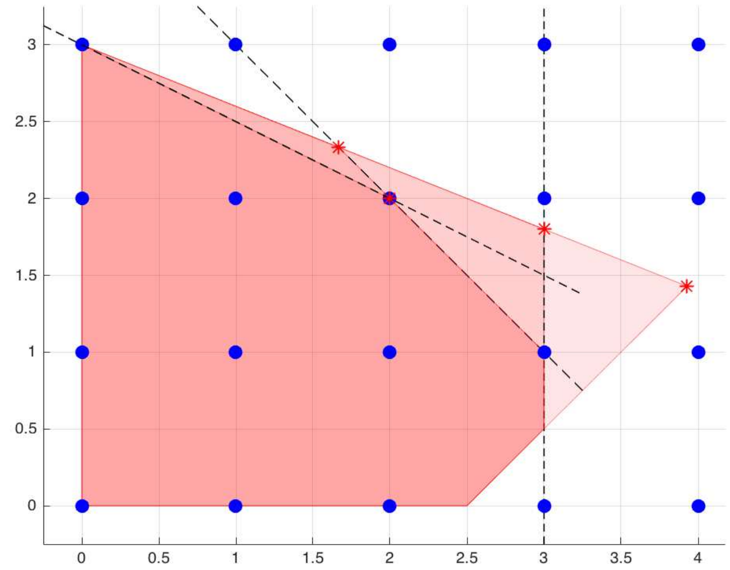
$$
Z^\star = \begin{cases} \min\limits_{x,\theta} & 2x_1 + 3x_2 + \sum_s p_s \theta_s \\ \text{s.t.} & x_1 + x_2 \leq 100 \\ & x_1, x_2 \geq 0 \\ & \theta_s \geq [h_1^s - (\alpha^s x_1 + 6x_2)]\pi_1 + \\ & \qquad [h_2^s - (3x_1 + \beta^s x_2)]\pi_2 \\ & \qquad \forall(\pi_1, \pi_2) \in \mathbf{XP}(\Pi_s), \forall s \end{cases}
$$

- For a LP with n variables, **at most n constraints** can be active at the optimal solution
- Do not need to generate all constraints upfront ! Add them dynamically as needed.

Cutting plane methods

- **Idea**: iteratively refine a feasible set or objective function by means of linear inequalities, termed cuts
- In our (Benders) reformulation

$$\theta_s \geq [h_1^s - (\alpha^s x_1 + 6x_2)]\pi_1 + [h_2^s - (3x_1 + \beta^s x_2)]\pi_2 \quad \forall(\pi_1, \pi_2) \in \mathbf{XP}(\Pi_s), \forall s$$



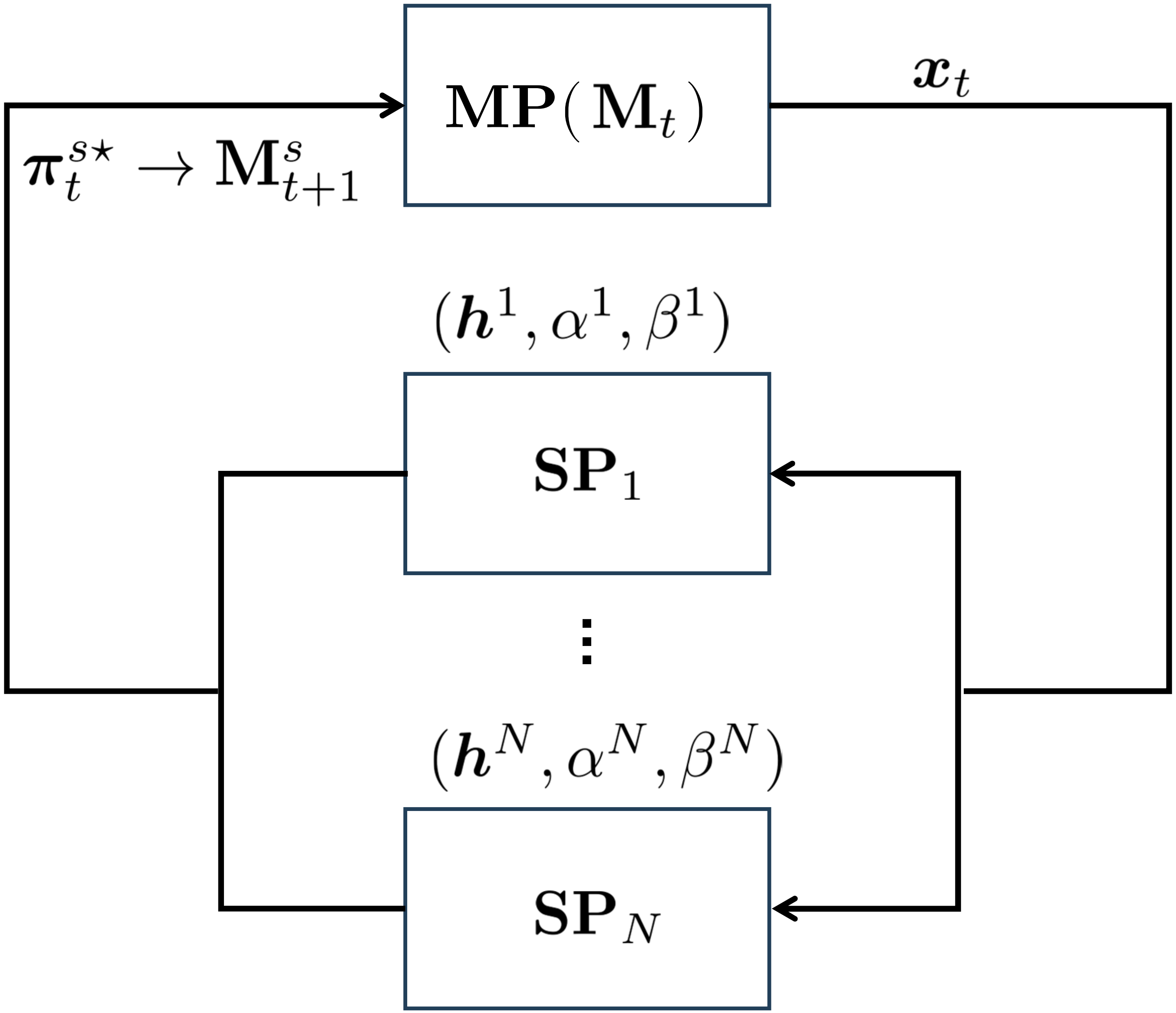
- are Benders (optimality) cuts
- **Benders decomposition**: specific method for obtaining the cuts for general 2-stage LP
- **L-shaped method**: specific instance of Benders decomposition when 2^{nd}-stage LP decomposable into scenarios

- Relax inequalities that define the scenario cost

$$
(\mathbf{MP}) : \mathrm{Z}_t = \begin{cases} \min\limits_{x,\theta} & 2x_1 + 3x_2 + \sum_s p_s\theta_s \\ \text{s.t.} & x_1 + x_2 \leq 100 \\ & x_1, x_2 \geq 0 \\ & \theta_s \geq [h_1^s - (\alpha^s x_1 + 6x_2)]\pi_1 + \\ & \qquad [h_2^s - (3x_1 + \beta^s x_2)]\pi_2 \\ & \qquad \forall(\pi_1, \pi_2) \in \mathbf{M}_t^s \subset \mathbf{XP}(\Pi_s), \forall s \end{cases}
$$

- Subproblem $(\mathbf{SP}_s)$: the dual of the 2nd-stage problem for a given x and for each scenario

Overall scheme



Bounds

CentraleSupélec

- Solution of the $\mathbf{MP}$ provides

  $\rightarrow$ lower bound $\mathrm{Z}_t \leq \mathrm{Z}^\star$

  $\rightarrow$ candidate feasible solution x_t

  $\rightarrow$ under-estimator $\theta_s^t \leq Q(x_t, \xi_s)$

- Solutions of all $\mathbf{SP}$ with input x_t provide

  $\rightarrow$ upper bound $\mathrm{Z}^{ub} = c^\top x_t + \sum_s p_s Q(x_t, \xi_s) \geq \mathrm{Z}^\star$

  $\rightarrow$ new dual vertex $\pi_{t+1}^{s\star}$, will added to $\mathbf{M}_{t+1}^s$ if not exist

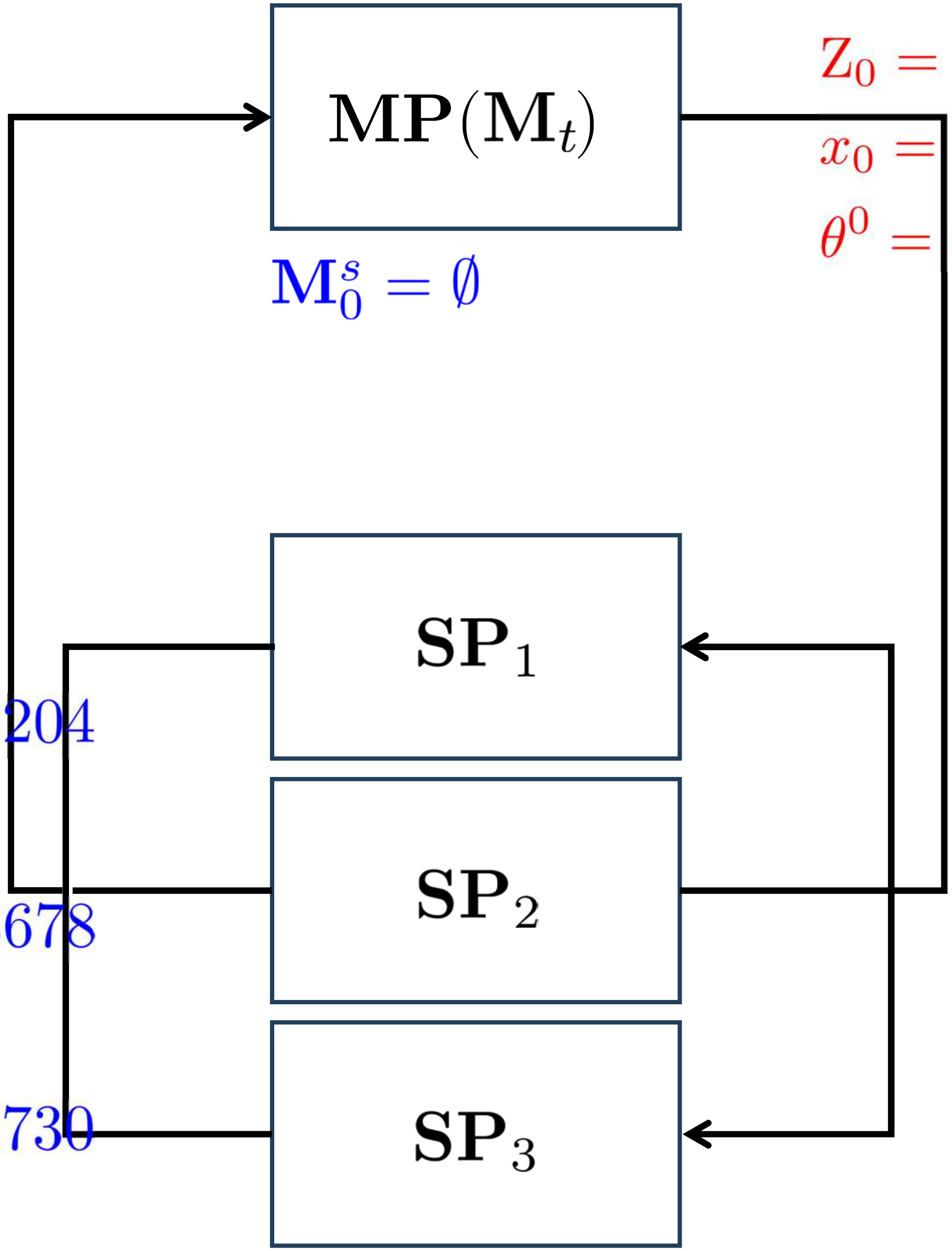
The production example

$$\xi_s = (h_1^s, h_2^s, \alpha^s, \beta^s) = \begin{cases} (180, 162, 2, 3) & p_1 = 0.6 \\ (210, 184, 2, 3) & p_2 = 0.2 \\ (150, 140, 2, 3) & p_3 = 0.2 \end{cases}$$

$$\mathbf{MP}(\mathbf{M}_t) : \mathrm{Z}_t = \begin{cases} \min\limits_{x,\theta} & 2x_1 + 3x_2 + \sum_s p_s \theta_s \\ \text{s.t.} & x_1 + x_2 \leq 100 \quad x_1, x_2 \geq 0 \\ & \theta_s \geq [h_1^s - (2x_1 + 6x_2)]\pi_1 + \\ & \qquad [h_2^s - (3x_1 + 3x_2)]\pi_2 \\ & \qquad \forall (\pi_1, \pi_2) \in \mathbf{M}_t^s, \forall s \end{cases}$$

$$\mathbf{SP}_s : Q(x_t, \xi_s) = \begin{cases} \max & [h_1^s - (2x_1 + 6x_2)]\pi_1 + \\ & [h_2^s - (3x_1 + 3x_2)]\pi_2 \\ \text{s.t.} & 0 \leq \pi_1 \leq 7, \\ & 0 \leq \pi_2 \leq 12 \end{cases}$$

Illustration



$\mathrm{Z}_0 = -\inf$

$x_0 = (0, 0)$

$\theta^0 = (-\inf, -\inf, -\inf)$

$\mathbf{M}_0^s = \emptyset$

$\mathrm{Z}^{ub} = 3204$

$\boldsymbol{\pi}_1^{1\star} = (7, 12)$

$Q(x_0, \xi_1) = 3204$

$\boldsymbol{\pi}_1^{2\star} = (7, 12)$

$Q(x_0, \xi_2) = 3678$

$\boldsymbol{\pi}_1^{3\star} = (7, 12)$

$Q(x_0, \xi_3) = 2730$

Illustration

$\mathbf{MP}(\mathbf{M}_t)$

$\mathrm{Z}_1 = -4296$

$x_1 = (0, 100)$

$\theta^1 = (-4596, -4122, -5070)$

$\mathbf{M}_1^s = \{(7, 12)\}, s = 1, 2, 3$

$$\theta_1 \geq 7 \cdot [180 - (2x_1 + 6x_2)] + 12 \cdot [162 - (3x_1 + 3x_2)]$$
$$\theta_2 \geq 7 \cdot [210 - (2x_1 + 6x_2)] + 12 \cdot [184 - (3x_1 + 3x_2)]$$
$$\theta_3 \geq 7 \cdot [150 - (2x_1 + 6x_2)] + 12 \cdot [140 - (3x_1 + 3x_2)]$$

$\mathrm{Z}^{ub} = 300$

$\boldsymbol{\pi}_2^{1\star} = (0, 0)$
$Q(x_1, \xi_1) = 0$

$\mathbf{SP}_1$

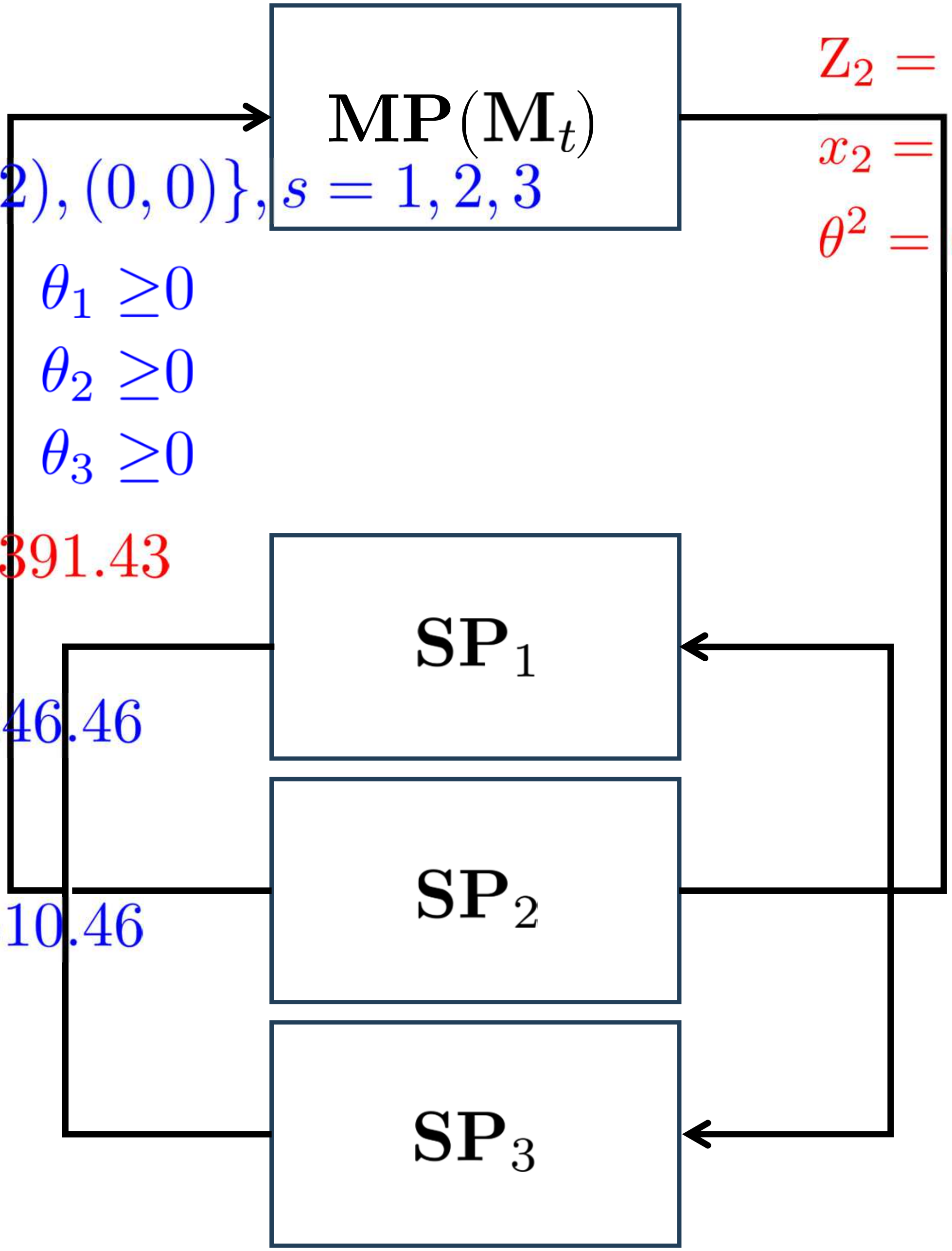
$\boldsymbol{\pi}_2^{2\star} = (0, 0)$
$Q(x_1, \xi_2) = 0$

$\mathbf{SP}_2$

$\boldsymbol{\pi}_2^{3\star} = (0, 0)$
$Q(x_1, \xi_3) = 0$

$\mathbf{SP}_3$

Illustration



$Z_2 = 141.46$

$x_2 = (0, 47.154)$

$\theta^2 = (0, 0, 0)$

$\mathbf{M}_2^s = \{(7, 12), (0, 0)\}, s = 1, 2, 3$

$\theta_1 \geq 0$

$\theta_2 \geq 0$

$\theta_3 \geq 0$

$Z^{ub} = 300 < 391.43$

$\boldsymbol{\pi}_3^{1\star} = (0, 12)$

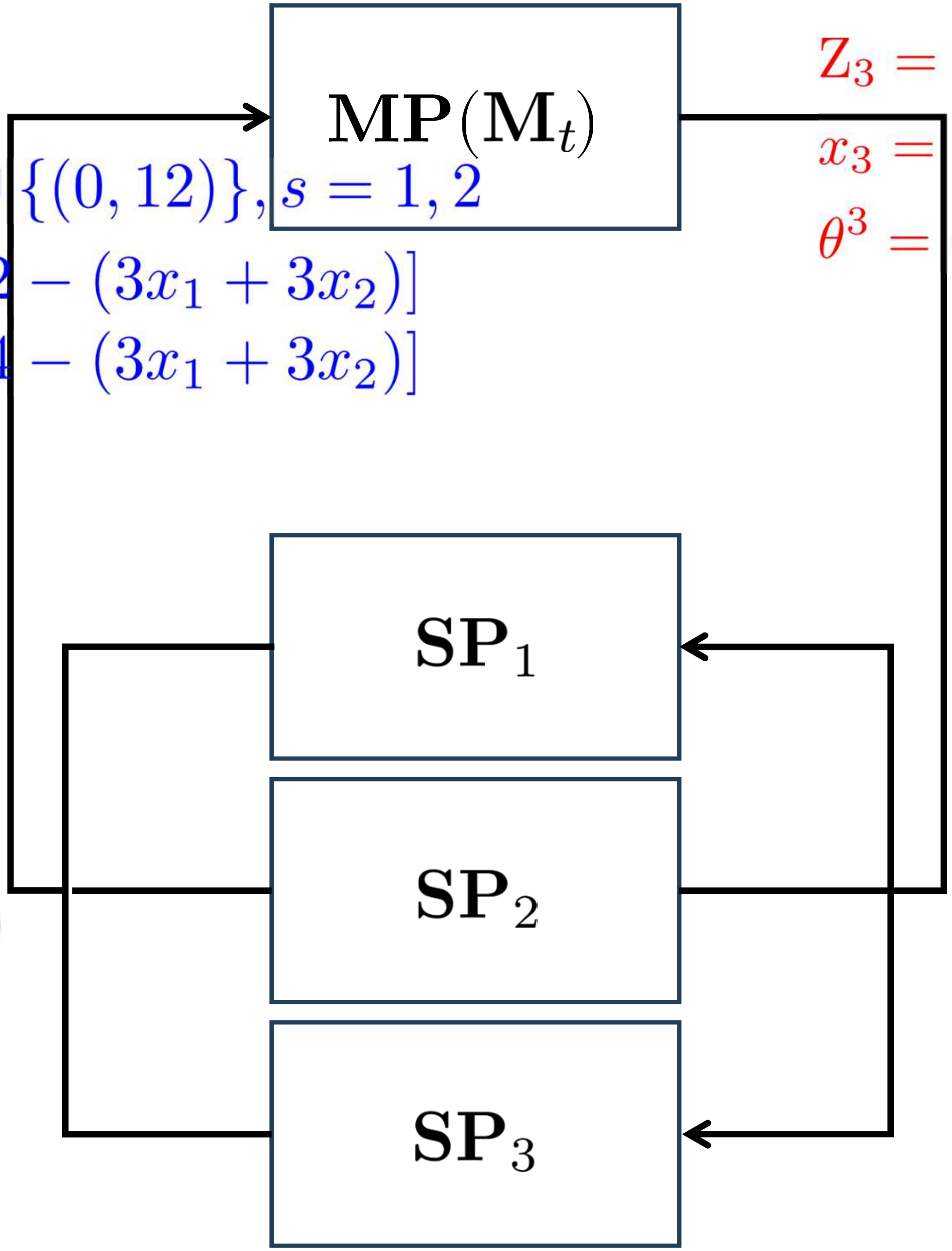
$Q(x_2, \xi_1) = 246.46$

$\boldsymbol{\pi}_3^{2\star} = (0, 12)$

$Q(x_2, \xi_2) = 510.46$

$\boldsymbol{\pi}_3^{3\star} = (0, 0)$

$Q(x_2, \xi_3) = 0$



$\mathrm{Z}_3 = 144.5$

$x_3 = (39.5, 21.83)$

$\theta^3 = (0, 0, 0)$

$\mathbf{M}_3^s = \mathbf{M}_2^s \cup \{(0, 12)\}, s = 1, 2$

$\theta_1 \geq 12 \cdot [162 - (3x_1 + 3x_2)]$

$\theta_2 \geq 12 \cdot [184 - (3x_1 + 3x_2)]$

$\mathrm{Z}^{ub} = 144.5$

$\boldsymbol{\pi}_4^{1\star} = (0, 0)$

$Q(x_2, \xi_1) = 0$

$\boldsymbol{\pi}_3^{2\star} = (7, 12)$

$Q(x_2, \xi_2) = 0$

$\boldsymbol{\pi}_3^{3\star} = (0, 0)$

$Q(x_2, \xi_3) = 0$

General case without RCR

- For a general 2-stage SLP with fixed recourse, but no RCR

$$(\mathbf{MP}) : \mathrm{Z}_t = \begin{cases} \min\limits_{x,\theta} & c^\top x + \sum_s p_s \theta_s \\ \text{s.t.} & Ax = b, \ x \geq 0 \\ & \theta_s \geq \pi^\top (h^s - T^s x), \forall \pi \in \mathbf{M}_t^s, \forall s \end{cases}$$

$$(\mathbf{SP}_s) : Q(x_t, \xi_s) \overset{\text{primal}}{=} \begin{cases} \min\limits_{y \in \mathbb{R}_+^m} & {q^s}^\top y \\ \text{s.t.} & Wy = h^s - T^s x \end{cases}$$

$$\overset{\text{dual}}{=} \begin{cases} \max & \pi^\top (h^s - T^s x_t) \\ \text{s.t.} & W^\top \pi \leq q^s \end{cases}$$

- Without RCR, the SP might be infeasible for some x

General case without RCR

- For what x is the subproblem feasible?
- Assume the dual is feasible (not depending on x)

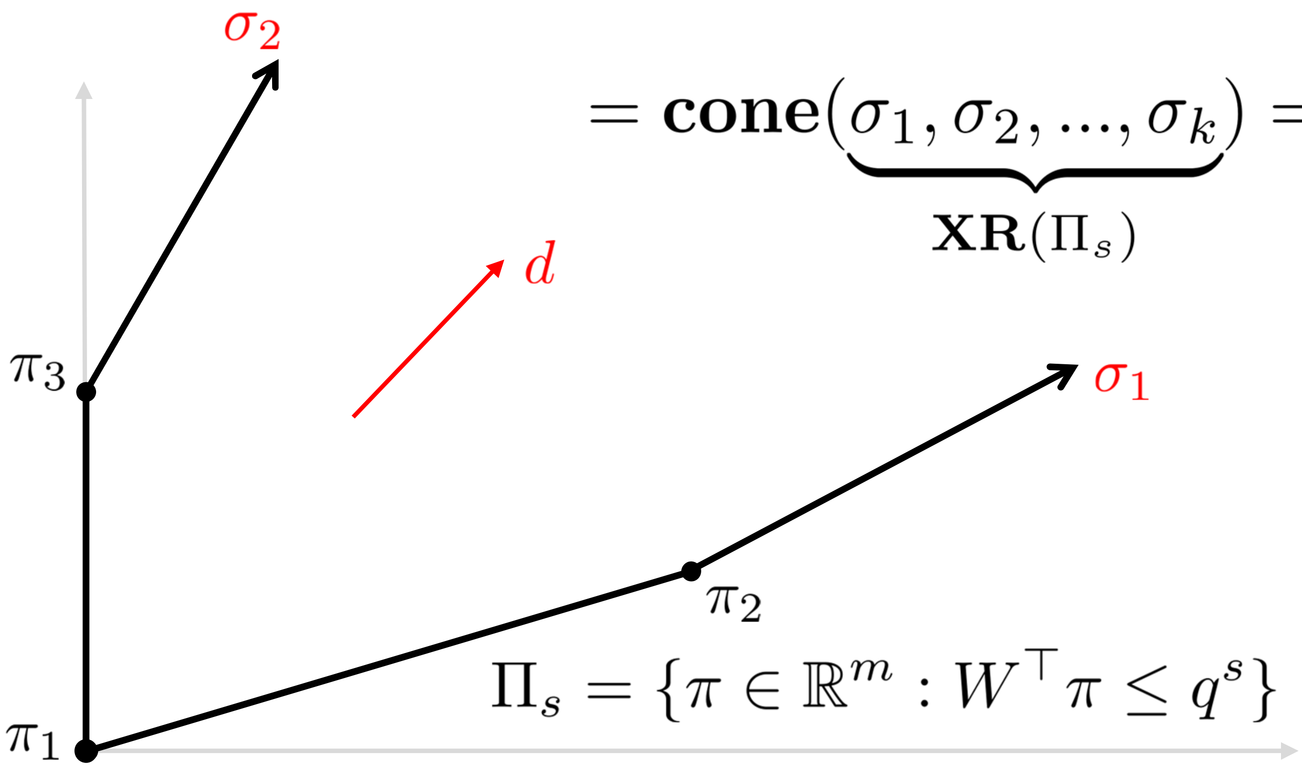
If the feasible dual is bounded $d^\star < +\infty$	$p^\star = d^\star$ $\Longleftrightarrow$	The bounded primal is feasible $p^\star < +\infty$

General case without RCR

$$\begin{cases} \text{max} & \pi^\top (h^s - T^s x) \\ \text{s.t.} & W^\top \pi \le q^s \end{cases} = \begin{cases} < +\infty & \boxed{d^\top (h^s - T^s x) \le 0} \\ +\infty & d^\top (h^s - T^s x) > 0 \end{cases}$$

$$\mathbf{rec}(\Pi_s) = \{d \in \mathbb{R}^m : W^\top d \le 0\}$$

$$= \mathbf{cone}(\underbrace{\sigma_1, \sigma_2, \dots, \sigma_k}_{\mathbf{XR}(\Pi_s)}) = \left\{ \sum_{i=1}^{k} \lambda_i \sigma_i : \lambda_i \ge 0 \right\}$$



$$\Pi_s = \{\pi \in \mathbb{R}^m : W^\top \pi \le q^s\}$$

General case without RCR

- Domain of $Q(x, \xi_s)$

$$Q(x, \xi_s) < +\infty \Leftrightarrow$$

$$x \in \left\{x \in \mathbb{R}^n : \sigma^\top (h^s - T^s x) \le 0,\ \forall \sigma \in \mathbf{XR}(\Pi_s)\right\}$$

- Modified Benders reformulation

$$Z^\star = \begin{cases} \min\limits_{x,\theta} & c^\top x + \sum_s p_s \theta_s \\ \text{s.t.} & Ax = b,\ x \ge 0 \\ & \boxed{\theta_s \ge \pi^\top (h^s - T^s x), \forall \pi \in \mathbf{XP}(\Pi_s), \forall s} \\ & \boxed{\sigma^\top (h^s - T^s x) \le 0,\ \forall \sigma \in \mathbf{XR}(\Pi_s), \forall s} \end{cases}$$

Optimality cuts

Feasibility cuts

- Again, can be solved using the Master-subproblem cutting planning strategy

L-shaped with feasibility cuts

- Master problem $(\mathbf{MP})_t$:

  $\rightarrow$ Replace full set $\mathbf{XP}(\Pi_s)$ with a subset $\mathbf{M}_t^s$ for each s

  $\rightarrow$ Replace full set $\mathbf{XR}(\Pi_s)$ with a subset $\mathbf{F}_t^s$ for each s

- Given a master solution (x_t, θ_s^t) solve each scenario s subproblem:

  $\rightarrow$ If subproblem s feasible add "optimality cut":

$$\mathbf{M}_{t+1}^s \leftarrow \mathbf{M}_t^s \cup \{\pi_s^\star\}$$

  $\rightarrow$ If subproblem s infeasible:

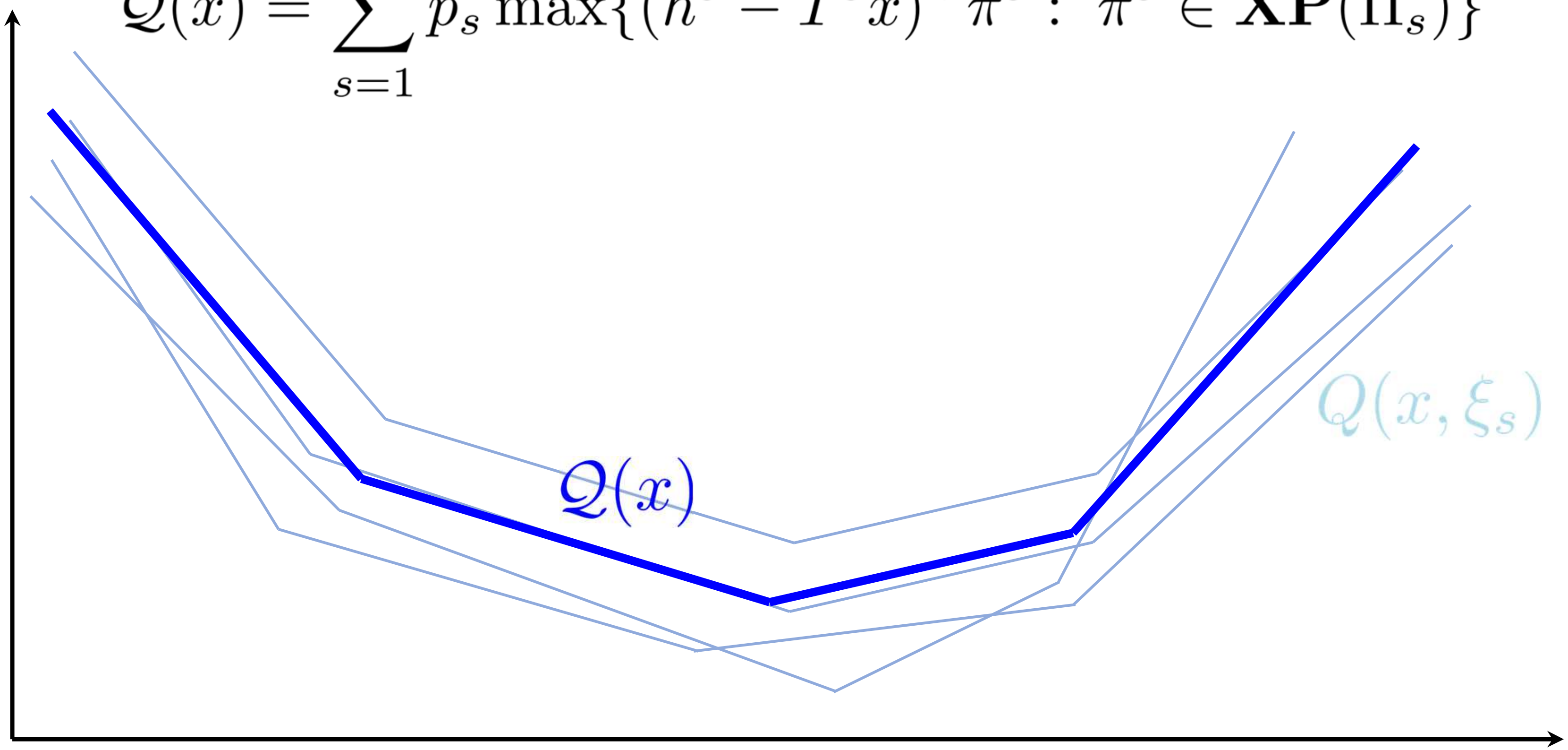
  - yield a dual extreme ray $\sigma_s \in \mathbf{XR}(\Pi_s)$ which gives

$$\sigma_s^\top (h_s - T_s x_t) > 0$$

  - add "feasibility cut": $\mathbf{F}_{t+1}^s \leftarrow \mathbf{F}_t^s \cup \{\sigma_s\}$

- We just saw the multi-cut version of the L-shaped
  $\rightarrow$ cuts are used to approximate the scenario value function
  $\rightarrow$ many cuts (at most N) may be added in each iteration
- **Alternative**: single-cut version

$$\mathcal{Q}(x) = \sum_{s=1}^{N} p_s \max\{(h^s - T^s x)^\top \pi^s \; : \; \pi^s \in \mathbf{XP}(\Pi_s)\}$$



Single-cut L-shaped

$$\mathcal{Q}(x) = \begin{cases} \min \Theta \\ \text{s.t. } \Theta \geq \sum_{s=1}^{N} p_s \max\{(h^s - T^s x)^\top \pi^s : \ \pi^s \in \mathbf{XP}(\Pi_s)\} \end{cases}$$

$$= \begin{cases} \min \Theta \\ \text{s.t. } \Theta \geq \sum_{s=1}^{N} p_s (h^s - T^s x)^\top \pi^s, \\ \qquad\qquad \forall (\pi^1, \ldots, \pi^N) \in \mathbf{XP}(\Pi_1) \times \ldots \times \mathbf{XP}(\Pi_N) \end{cases}$$

- Huge number of constraints but less additional variables
- For cutting plane algorithm, only need to be able to efficiently find a violated constraint

Single-cut master problem

- Master problem $(\mathbf{MP})_t$ after adding t optimality cuts

$$
(\mathbf{MP})_t : \mathrm{Z}_t = \begin{cases} \min\limits_{x,\Theta} & c^\top x + \Theta \\ \text{s.t.} & Ax = b,\ x \geq 0 \\ & \sigma^\top (h^s - T^s x) \leq 0,\ \forall \sigma \in \mathbf{F}_t^s, \forall s \\ & \Theta \geq \sum_{s=1}^N p_s (h^s - T^s x)^\top \pi^s, \\ & \forall (\pi^1, \ldots, \pi^N) \in \mathbf{M}_t^N \end{cases}
$$

Single-cut vs. multi-cut

- **Single-cut**

  $\rightarrow$ fewer variables and optimality cuts $\Rightarrow$ master problem typically solved fasters

  $\rightarrow$ must solve all subproblems to generate an optimality cut

- **Multi-cut**

  $\rightarrow$ many cuts added per iteration $\Rightarrow$ typically converges in fewer iterations

  $\rightarrow$ master problem may become large and slow to solve

Outline

- Benders decomposition/L-Shaped method
- Lagrangian dual decomposition

Production planning example

- Splitted formulation: use scenario (local) versions of 1st stage (complicating) variable x

$$
\begin{aligned}
Z^{\star} = \min \quad & \sum_{s=1}^{N} p_s[2x_1^s + 3x_2^s + 7y_1^s + 12y_2^s)] \\
\text{s.t.} \quad & x_1^s + x_2^s \leq 100, \; x_1^s, x_2^s \geq 0, \; \forall s \\
& \alpha^s x_1^s + 6x_2^s + y_1^s \geq h_1^s, \; \forall s \\
& 3x_1^s + \beta^s x_2^s + y_2^s \geq h_2^s, \; \forall s \\
& y_1^s, y_2^s \geq 0, , \; \forall s \\
& \boxed{x_1^1 = x_1^2 = \ldots = x_1^N, x_2^1 = x_2^2 = \ldots = x_2^N}
\end{aligned}
$$

Production planning example

- Splitted formulation: use scenario (local) versions of 1[st] stage (complicating) variable x

$$
\begin{aligned}
Z^\star = \min \quad & \sum_{s=1}^{N} p_s[2x_1^s + 3x_2^s + 7y_1^s + 12y_2^s)] \\
\text{s.t.} \quad & x_1^s + x_2^s \le 100,\ x_1^s, x_2^s \ge 0,\ \forall s \\
& \alpha^s x_1^s + 6x_2^s + y_1^s \ge h_1^s,\ \forall s \\
& 3x_1^s + \beta^s x_2^s + y_2^s \ge h_2^s,\ \forall s \\
& y_1^s, y_2^s \ge 0, ,\ \forall s \\
& \boxed{x_1^1 = x_1^2 = \ldots = x_1^N, x_2^1 = x_2^2 = \ldots = x_2^N}
\end{aligned}
$$

$$
x_1^s = \sum_{s'=1}^{N} p_{s'} x_1^{s'},\ s = 1, \ldots, N \qquad x_2^s = \sum_{s'=1}^{N} p_{s'} x_2^{s'},\ s = 1, \ldots, N
$$

Splitted formulation of two-stage SP

$$
\begin{aligned}
\min_{x_s, y_s} \quad & \sum_{s=1}^{N} p_s(c^\top x_s + q_s^\top y_s) && \Rightarrow \quad \sum_{s=1}^{N} p_s f(x_s, \xi_s, y_s) \\
\text{s.t.} \quad & Ax_s = b, \quad \forall s \\
& x_s \geq 0, \quad \forall s \\
& W y_s + T_s x = h_s, \quad \forall s && \Rightarrow \quad g_s(x_s, \xi_s, y_s) \leq 0 \; \forall s \\
& && \quad x_s \in X_s, y_s \in Y_s \; \forall s \\
& y_s \geq 0, \quad \forall s \\
& x_1 = \cdots = x_N
\end{aligned}
$$

- Another way to write the nonanticipativity (consensus) constraints

$$
x_s = \sum_{i=1}^{N} p_i x_i, \; s = 1, \ldots, N
$$

Splitted formulation of two-stage SP

$$\min_{\{x_s \in X_s, y_s \in Y_s\}_{s=1}^{S}} \quad \sum_{s=1}^{N} p_s f(x_s, \xi_s, y_s)$$

$$\text{s.t.} \quad x_s = \sum_{i=1}^{N} p_i x_i, \ \forall s \qquad \lambda_s$$

$$\begin{aligned}
L(x,\lambda) &= \sum_{s=1}^{N} p_s f(x_s, \xi_s, y_s) + \sum_{s=1}^{N} p_s \lambda_s \left(x_s - \sum_{i=1}^{N} p_i x_i\right) \\
&= \sum_{s=1}^{N} p_s f(x_s, \xi_s, y_s) + \sum_{s=1}^{N} p_s \lambda_s x_s - \sum_{s=1}^{N} \sum_{i=1}^{N} p_s \lambda_s p_i x_i \\
&= \sum_{s=1}^{N} p_s f(x_s, \xi_s, y_s) + \sum_{s=1}^{N} p_s \left(\lambda_s - \sum_{i=1}^{N} p_i \lambda_i\right) x_s
\end{aligned}$$

Dualization of nonanticipativity constraint

$$L(x,y,\lambda) = \sum_{s=1}^{N} p_s f(x_s,\xi_s,y_s) + \sum_{s=1}^{N} p_s \left(\lambda_s - \sum_{i=1}^{N} p_i \lambda_i \right) x_s$$

- Shift the multiplier λ by a constant vector does not change the value of the Lagrangian at optimality $\Rightarrow$ thus, assume

$$\mathbf{E}[\lambda] = \mathbf{p}\lambda = \sum_{i=1}^{S} p_i \lambda_i = 0$$

- Dual problem with respect to the nonanticipativity constraints

$$\max_{\lambda} \quad \left\{ D(\lambda) \; = \inf_{\{x_s \in X_s, y_s \in Y_s\}_{s=1}^{S}} L(x,y,\lambda) \right\}$$
$$\text{s.t.} \quad \mathbf{p}\lambda = 0$$

Dualization of nonanticipativity constraint

$$
\begin{aligned}
\max_{\lambda} \quad & \min_{\{x_s \in X_s, y_s \in Y_s\}_{s=1}^{S}} \sum_{s=1}^{N} p_s f(x_s, \xi_s, y_s) + \sum_{s=1}^{N} p_s \lambda_s x_s \\
\text{s.t.} \quad & \mathbf{p}\lambda = 0
\end{aligned}
$$

- The inner minimization, for λ given, can decompose scenario by scenario, by solving S deterministic problems

$$
\begin{aligned}
\min_{x_s, y_s} \quad & f(x_s, \xi_s, y_s) + \lambda_s x_s \\
\text{s.t.} \quad & g_s(x_s, \xi_s, y_s) \leq 0
\end{aligned}
$$

Price of information

- By weak duality theory, any feasible $\boldsymbol{\lambda}$ will give a lower bound on the 2-stage SP problem, computed by the augmented anticipative scenario problem:

$$\begin{aligned} \min_{x_s, y_s} \quad & f(x_s, \xi_s, y_s) + \lambda_s x_s \\ \text{s.t.} \quad & g_s(x_s, \xi_s, y_s) \leq 0 \end{aligned}$$

- $\lambda = 0$ lead to the anticipative lower-bound
- If problem is convex, and under some qualification assumptions, there exists an optimal λ^*, called the price of information, such that the lower bound is tight.
- E.g., linear 2-stage SP problems

Progressive hedging algorithm

The PH algorithm builds on the dual decomposition in the following way:

1) Set a price of information $\{\lambda_s\}_{s=1}^N$ such that $\mathbf{p}\lambda = 0$
2) For each scenario s, solve

$$\begin{aligned} \min_{x_s, y_s} \quad & f(x_s, \xi_s, y_s) + \lambda_s x_s \\ \text{s.t.} \quad & g_s(x_s, \xi_s, y_s) \leq 0 \end{aligned}$$

3) Compute the mean first decision $\bar{x} = \sum_{s=1}^N p_s x_s$
4) Update the price of information with

$$\lambda_s = \lambda_s + \rho(x_s - \bar{x})$$

5) If $\sum_{s=1}^N p_s \|x_s - \bar{x}\| \geq \epsilon_{ph}$ go back to step 2), otherwise terminate

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$$\begin{aligned} \min_{x_s, y_s} \quad & f(x_s, \xi_s, y_s) + \lambda_s x_s + \frac{\rho}{2}\|x_s - \bar{x}\|^2 \\ \text{s.t.} \quad & g_s(x_s, \xi_s, y_s) \leq 0 \end{aligned}$$

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Rockafellar and Wets (1991)

Convergence of PH algorithm

CentraleSupélec

Theorem (Rockafellar and Wets, 1991)

If $f(x,\xi,y)$ and $g(x,\xi,y)$ are convex l.s.c. functions in (x,y) for all ξ, and that for all s, there exist (x_s, y_s) such that

$$f(x_s, \xi_s, y_s) \leq +\infty$$
$$g_s(x_s, \xi_s, y_s) \leq 0$$

Then, the PH algorithm converges toward an optimal primal solution, and the price of information converges toward an optimal price of information.

Moreover, it shows that

$$\epsilon_k = \sqrt{\|(x^k, y^k) - (x^\star, y^\star)\|_2^2 + \frac{1}{\rho^2}\|\lambda^k - \lambda^\star\|_2^2}$$

is a decreasing sequence

Bounds in progressive hedging

- At any iteration of the PH algorithm, we have a collection of primal solution $\{(x_s, y_s)\}_{s=1}^{S}$, and a price of information $\{\lambda_s\}_{s=1}^{S}$

- We have a lower bound on the value of the SP problem given by

$$\mathbf{LB}^{ph} = \sum_{s=1}^{N} p_s[f(x_s, \xi_s, y_s) + \lambda_s x_s]$$

- and an upper bound given by

$$\mathbf{UB}^{ph} = \sum_{s=1}^{N} p_s f(\bar{x}, \xi_s, y_s(\bar{x}))$$

Comparison of L-shaped and PH algorithm

	Dual decomposition (PH)	L-Shaped Benders
Problems	Convex Lower bound if nonconvex	2^{nd} stage LP, 1^{st} stage can be MIP
Convergence	Asymptotic	finite
Complexity	fixed: S deterministic problems Simple (projected subgradient) update for the dual price	Increasing number of cuts, and thus complexity, for master problem, fixed for subproblems

Exercise

- Implement the L-shaped method to for your projects
- Implementation note on the extreme rays:

  → Pyomo does not directly support extreme ray extraction

  → Solvers like CPLEX and Gurobi can detect unboundedness and provide extreme rays through its native Python API

yiping.fang@centralesupelec.fr